



Cambridge IGCSE™

MATHEMATICS

0580/43

Paper 4 (Extended)

October/November 2022

MARK SCHEME

Maximum Mark: 130

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **13** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

PUBLISHED**GENERIC MARKING PRINCIPLE 6:**

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles

1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfw	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	60.9 or 60.86 to 60.87	1	
1(a)(ii)	375	2	M1 for $\frac{250}{12} [\times 18]$ oe
1(a)(iii)	30 nfw	3	M1 for $2200 \div 800 [\times 12]$ oe M1 for $1500 \div 600 [\times 12]$ oe
1(b)(i)	1.92	2	M1 for $k \times \left(1 + \frac{25}{100}\right) = 2.4[0]$ oe or better
1(b)(ii)	43.75 or $43\frac{3}{4}$	3	M2 for $\left(\left(1 + \frac{25}{100}\right) \times \left(1 + \frac{15}{100}\right) [-1]\right) [\times 100]$ oe or $\left(1 + \frac{25}{100}\right) \times \left(1 + \frac{15}{100}\right) \times 100 [-100]$ or for $\frac{2.40 \times \left(1 + \frac{15}{100}\right)}{\text{their (b)(i)}} \times 100 [-100]$ oe or M1 for $2.40 \times \left(1 + \frac{15}{100}\right)$ or $\left(1 + \frac{25}{100}\right) \times \left(1 + \frac{15}{100}\right)$ oe
1(c)	18 nfw	3	M2 for $\frac{200 \text{ to } 210}{11.5 - 0.25}$ or $\frac{200 + 5}{11 \text{ to } 11.5}$ oe or M1 for $200 + 5$, $200 - 5$, $11.5 + 0.25$ or $11.5 - 0.25$

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Question	Answer	Marks	Partial Marks
2(a)(i)	p^{14} final answer	1	
2(a)(ii)	$6m^4$ final answer	2	B1 for $6m^k$ or km^4 in final answer or correct answer seen and spoilt
2(a)(iii)	$\frac{4}{3x^3y^9}$ or $\frac{4x^{-3}y^{-9}}{3}$ final answer	3	B2 for correct answer seen and spoilt or 2 correct elements in final answer or B1 for one of $\frac{4}{3}$ or $\frac{3}{4}$ oe or x^3 or y^9 seen
2(b)	3, 12, 27	2	B1 for 12 or 27
2(c)(i)	$3n + 10$ oe final answer	2	B1 for $3n + k$ oe or $jn + 10$ oe ($j \neq 0$) or for correct expression shown in working and then spoilt
2(c)(ii)	$2n^3 + 1$ oe final answer	2	B1 for 3rd diff = 12 (both needed) or for cubic answer or for correct expression shown in working and then spoilt
2(d)	38	3	M2 for $3x = 4 \times 23 + 22$ or M1 for $3x - 22 = 4 \times 23$ or for $\frac{3x}{4} = 23 + \frac{22}{4}$ oe
2(e)	$\frac{-8 \pm \sqrt{8^2 - 4(3)(-20)}}{2 \times 3}$ or $\frac{-8}{2 \times 3} \pm \sqrt{\frac{8^2}{4 \times 3^2} - \frac{(-20)}{3}}$ or better	B2	B1 for $\sqrt{8^2 - 4(3)(-20)}$ oe or $\frac{-8 + \sqrt{q}}{2 \times 3}$ oe or $\frac{-8 - \sqrt{q}}{2 \times 3}$ oe or both
	– 4.24, 1.57 final answers	B2	B1 for each If B0 , SC1 for answers – 4.2 or –4.23 or –4.240 to – 4.239 and 1.6 or 1.572 to 1.573 or – 4.24 and 1.57 seen in working or for –1.57 and 4.24 as final answer

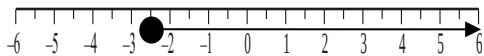
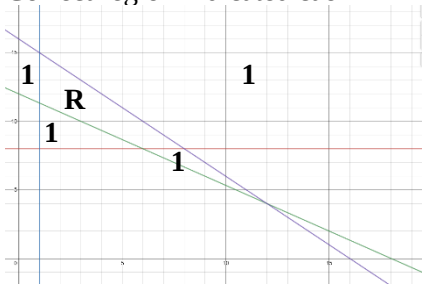
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Question	Answer	Marks	Partial Marks
3(a)	Correct histogram	3	B1 for each correct block If 0 scored, SC1 for two of $\frac{28}{15}$, $\frac{33}{20}$, $\frac{13}{10}$ or 1.87 or 1.866 to 1.867, 1.65, 1.3
3(b)	38.65	4	M1 for 12.5, 20, 32.5, 50, 65 soi M1 for $\sum fx$ where x is in the correct interval including boundaries M1dep for $\sum fx \div 100$
4(a)	Triangle drawn at (1, – 5), (1, – 7), (5, – 5)	2	B1 for reflection in any horizontal line If 0 scored, SC1 for reflection in $x = -2$
4(b)	Triangle drawn at (– 2, 0), (– 2, – 1), (0, – 1)	2	B1 for correct size and orientation but wrong position
4(c)	Rotation 90 [anticlockwise] oe [centre] (– 1, 0)	3	B1 for each

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Question	Answer	Marks	Partial Marks
5(a)	$\left(\frac{(36+50) \times 40}{2}\right) \times 120$ oe or $\left(\frac{(0.36+0.5) \times 0.4}{2}\right) \times 1.2$ oe	M2	M1 for $\frac{(36+50) \times 40}{2}$ oe or $\frac{(0.36+0.5) \times 0.4}{2}$ oe
	206400 ÷ 1000 = 206.4 or 0.2064 × 1000 = 206.4 nfw	A1	Must see an explicit conversion
5(b)	5 [minutes] 44 seconds	3	B2 for 344 [seconds] oe 5.73...[mins] or M1 for figs206.4 ÷ figs 6 oe
5(c)(i)	28[.0] or 27.96 to 27.97	3	M2 for $[r^2 =] \frac{\text{figs } 2064}{(\text{figs } 84)\pi}$ or M1 for $\pi r^2 \times \text{figs } 84 = \text{figs } 2064$
5(c)(ii)	140 cao	2	M1 for $0.6h = 84$ oe ALT method M1 for $\pi \times (\text{their } \mathbf{(c)(i)})^2 \times h = \text{figs } 206400 \div 0.6$ oe
5(d)	128 or 127.7 to 127.8	4	B3 for $40^2 + 120^2 + 18^2$ oe OR B1 for horizontal length 18 soi M1 for any correct attempt at 2-dimensional Pythagoras' $18^2 + 120^2$, $120^2 + 40^2$, $18^2 + 40^2$

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Question	Answer	Marks	Partial Marks
6(a)(i)	38	2	M1 for $5 \times 3^2 - 7$ oe
6(a)(ii)	$[\pm]\sqrt{\frac{P+7}{5}}$ oe final answer	3	M1 for $P + 7 = 5k^2$ or $\frac{P}{5} = k^2 - \frac{7}{5}$ M1 for $k^2 = \dots\dots$ FT <i>their</i> first step M1 for square root to final answer Max M2 for incorrect answer
6(b)(i)	$x \geq -2.5$ final answer	2	M1 for $-4x \leq 7 + 3$ or better
6(b)(ii)		1	FT <i>their</i> inequality in (b)(i)
6(c)(i)	$x = 2$ broken line	B1	
	$y = 32 - x$ solid line	B1	
	$2x + 3y = 72$ solid line	B2	B1 for line passing through (0, 24) or (36, 0)
	Correct region indicated cao 	B2	B1 for region satisfying 3 of the inequalities
6(c)(ii)	(16, 16)	2	M1 for substitution into $2x + y$ for any integer point in <i>their</i> region

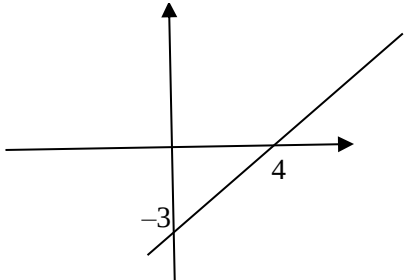
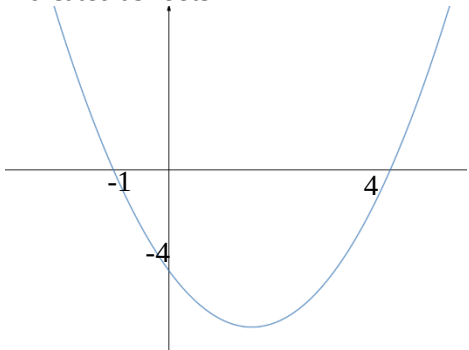
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Question	Answer	Marks	Partial Marks
7(a)(i)	$\frac{1}{15}$ oe	3	M2 for $2 \times \frac{1}{6} \times \frac{1}{5}$ oe or M1 for $\frac{1}{6} \times \frac{1}{5}$ oe or list or indication of 2 correct pairs If 0 scored, SC1 for answer $\frac{1}{18}$
7(a)(ii)	$\frac{7}{15}$ oe	3	M2 for $\left(\frac{4}{6} \times \frac{3}{5}\right) + 2\left(\frac{1}{6} \times \frac{1}{5}\right)$ oe or $14\left(\frac{1}{6} \times \frac{1}{5}\right)$ oe or $1 - 2\left(\frac{2}{6} \times \frac{4}{5}\right)$ or M1 for $\left(\frac{4}{6} \times \frac{3}{5}\right)$ or $2\left(\frac{1}{6} \times \frac{1}{5}\right)$ oe or $2\left(\frac{2}{6} \times \frac{4}{5}\right)$ or correct identification of 14 pairs If 0 scored, SC1 for answer $\frac{5}{9}$
7(b)	$\frac{1}{10}$ oe nfw	4	M3 for $6\left(\frac{1}{6} \times \frac{1}{5} \times \frac{1}{4}\right) + 6\left(\frac{1}{6} \times \frac{1}{5} \times \frac{1}{4}\right)$ oe or M2 for $6\left(\frac{1}{6} \times \frac{1}{5} \times \frac{1}{4}\right)$ oe or $2\left(\frac{1}{6} \times \frac{1}{5} \times \frac{1}{4}\right)$ oe or M1 for $k\left(\frac{1}{6} \times \frac{1}{5} \times \frac{1}{4}\right)$ where k is an integer and $1 \leq k \leq 12$ but not $k = 2$ or $k = 6$ or identifies $-2, 2$ and 5 or $-3, 3$ and 5 as the 3 cards needed If 0 scored, SC1 for answer $\frac{1}{18}$

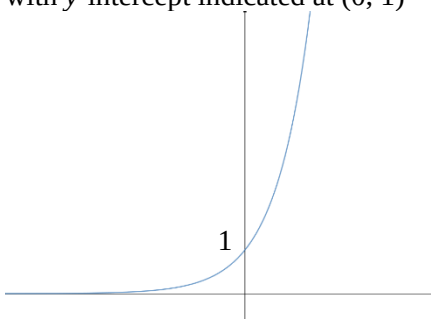
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Question	Answer	Marks	Partial Marks
8(a)	$[\cos B =] \frac{9.5^2 + 7.7^2 - 10^2}{2 \times 9.5 \times 7.7}$ oe	M2	M1 for $10^2 = 9.5^2 + 7.7^2 - 2 \times 9.5 \times 7.7 \cos B$ oe or better
	70.206 to 70.207 or 70.21 to 70.22	A2	A1 for $\frac{2477}{7315}$ oe or 0.339 or 0.3386....
8(b)(i)	140.4	1	
8(b)(ii)	19.8	1	FT $(180 - \text{their (b)(i)}) \div 2$
8(b)(iii)	70.2	1	FT $90 - \text{their (b)(ii)}$
8(c)	5.31 or 5.314 to 5.315	3	M2 for $\frac{5}{\cos \text{their (b)(ii)}}$ oe or M1 for $\frac{5}{r} = \cos(\text{their (b)(ii)})$ oe
8(d)	38.8 or 38.9 or 38.78 to 38.85	4	M3 for $\frac{0.5 \times 9.5 \times 7.7 \times \sin 70.2}{\pi \times (\text{their (c)})^2} [\times 100]$ OR M1 for $0.5 \times 9.5 \times 7.7 \times \sin 70.2$ M1 for $\pi \times (\text{their (c)})^2$

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Question	Answer	Marks	Partial Marks
9(a)(i)	Correct sketch of $3x - 4y = 12$ with $y = -3$ and $x = 4$ indicated on axes 	2	B1 for line with positive gradient
9(a)(ii)	Correct sketch of $y = x^2 - 3x - 4$ with $(0, -4)$ indicated as y – intercept and $x = -1$ and $x = 4$ indicated as roots  Minimum in fourth quadrant, not at $x = 0$	4	B3 for correct sketch with one value omitted or incorrect or for a poor sketch with all 3 intercepts correct. or B2 for roots $x = -1$ and $x = 4$ soi with no extra roots or for correct shape with $y = -4$ indicated or B1 for correct shape or for $(x - 4)(x + 1)$ shown or for incorrect sketch with $(0, -4)$ indicated as y – intercept

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Question	Answer	Marks	Partial Marks
9(a)(iii)	Correct sketch of $y = 6^x$ with y-intercept indicated at (0, 1) 	2	B1 for increasing exponential graph seen on both sides of the y-axis.
9(b)(i)	$8 - 4x^2$ [+ 0]	2	B1 for two terms correct and one extra incorrect term or for one of two terms correct or for correct answer seen and spoilt
9(b)(ii)	4	2	M1 for substitution of $x = -1$ into <i>their</i> (b)(i)
9(b)(iii)	(3, -7) and (-3, 17)	5	B4 for (3, -7) or (-3, 17) or B3 for $x = \pm 3$ or M2 for $x^2 = 9$ or $k(x - 3)(x + 3) = 0$ oe or for correct method for solving <i>their</i> (b)(i) = - 28 or M1 for <i>their</i> (b)(i) = - 28

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Question	Answer	Marks	Partial Marks
10(a)(i)	2a drawn correctly with direction arrow	1	
10(a)(ii)	a – b drawn correctly with direction arrow	2	B1 for $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$ seen or implied or M1 for correctly drawing <i>their</i> a – b with an arrow
10(b)(i)(a)	$q + \frac{3}{4}p$ final answer	1	
10(b)(i)(b)	$q - \frac{1}{4}p$ final answer	2	M1 for a correct route
10(b)(i)(c)	$\frac{13}{24}p - \frac{2}{3}q$ final answer	3	M2 for $\frac{3}{8}p - \frac{2}{3}$ (<i>their (b)(i)(b)</i>) oe or for $-\frac{3}{8}p - q + p + \frac{1}{3}$ (<i>their (b)(i)(b)</i>) oe or M1 for a correct route or for $\overrightarrow{BN} = -\frac{2}{3}$ (<i>their (b)(i)(b)</i>) or $\overrightarrow{AN} = \frac{1}{3}$ (<i>their (b)(i)(b)</i>) or final answer $kp - \frac{2}{3}q$ oe or $\frac{13}{24}p - kq$ oe
10(b)(ii)	$\frac{19}{16}p$ oe final answer	2	$\overrightarrow{AG} = \frac{3}{8}p \div 2$ soi M1 for or for answer kp oe